

ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 10: PLL Noise Transfer Functions

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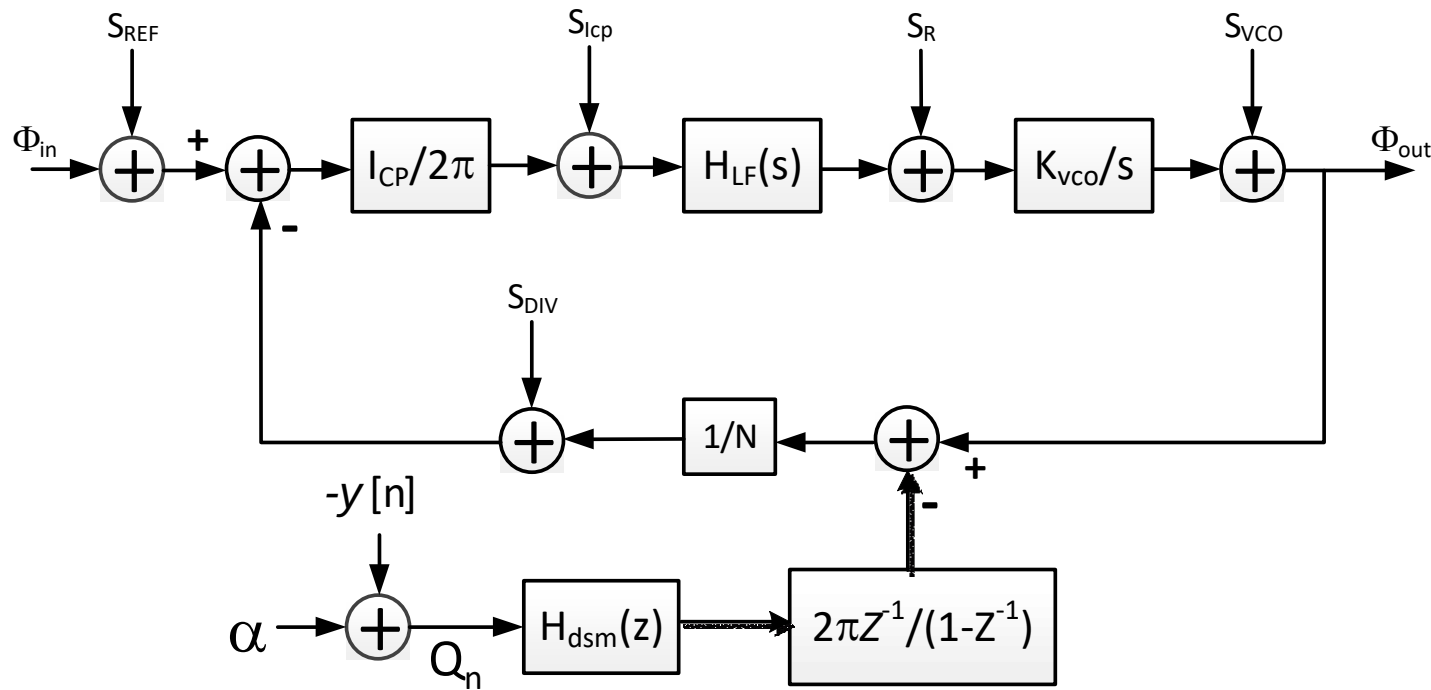


Outline

- PLL noise transfer functions of
 - Reference
 - PFD and CP
 - Loop filter
 - VCO
 - N-divider
 - Delta sigma modulator (DSM)
- Composite PLL noise profile and IPN calculation
- DSM noise folding
- PLL loop bandwidth tradeoff



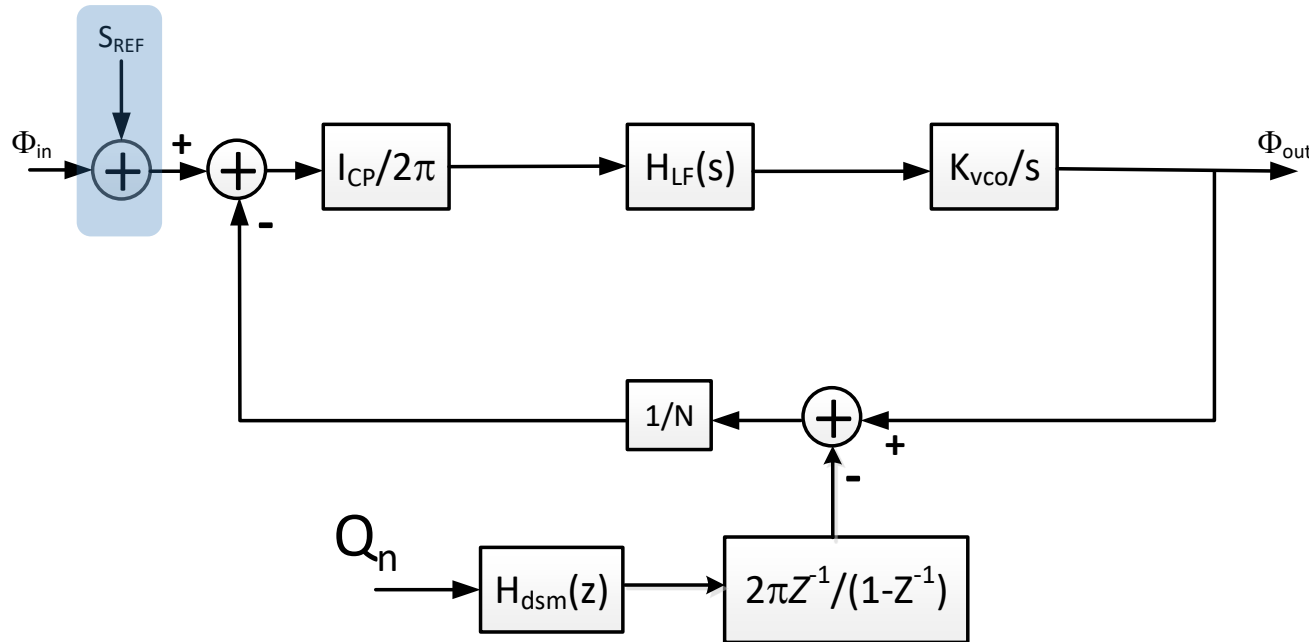
Noise Sources in PLL



- Reference signal noise S_{REF}
- PFD and charge pump S_{lcp}
- Loop filter resistor noise S_R
- VCO phase noise S_{VCO}
- N divider noise S_{DIV}
- Delta sigma quantization noise Q_n



Reference Noise to PLL Output



- Feedforward path transfer function

$$H_{FF}(s) = \frac{I_{CP}K_{VCO}}{2\pi s} H_{LF}(s)$$

- Feedback path transfer function

$$H_{FB}(s) = \frac{1}{N}$$

- Open loop transfer function

$$H_{OL}(s) = H_{FF} \cdot H_{FB} = \frac{I_{CP}K_{VCO}}{2\pi N s} H_{LF}(s)$$

- Transfer function from S_{REF} to Φ_{out} :

$$(S_{REF} - H_{FB}\Phi_{out})H_{FF} = \Phi_{out}$$

$$H_{REF}(s) = \frac{\Phi_{out}}{S_{REF}} = \frac{H_{FF}}{1 + H_{OL}} = \frac{N \cdot H_{OL}}{1 + H_{OL}}$$

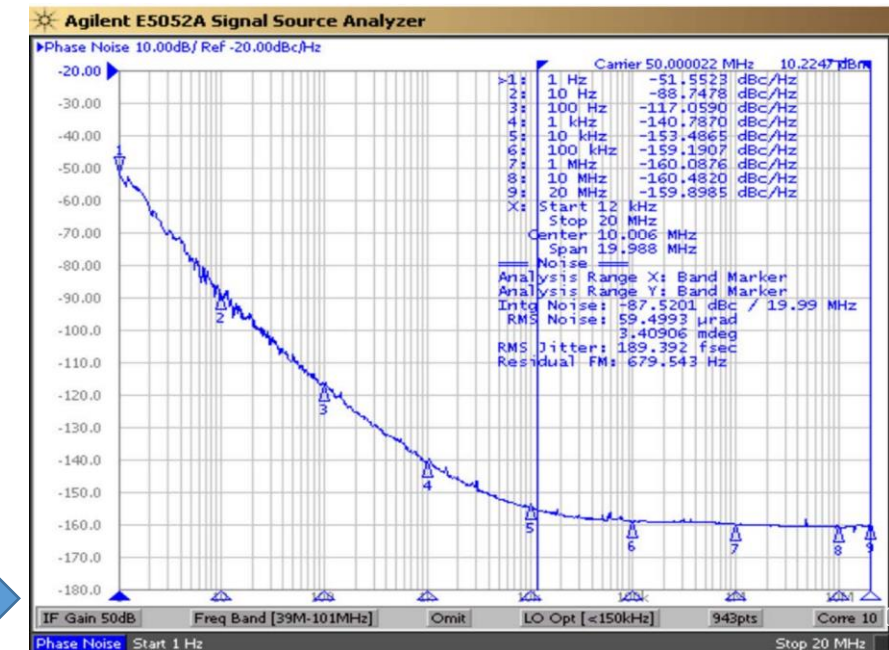
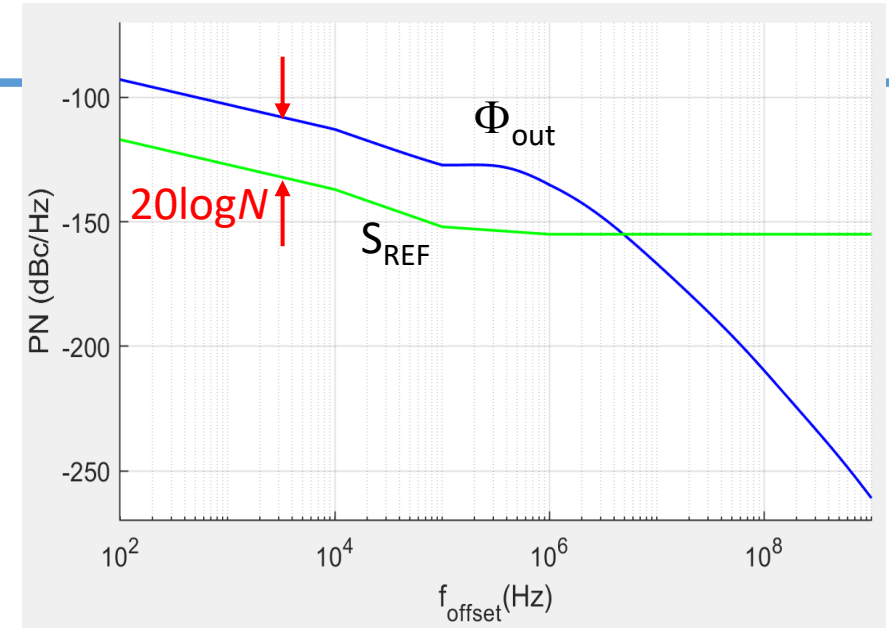


Reference Noise to PLL Output

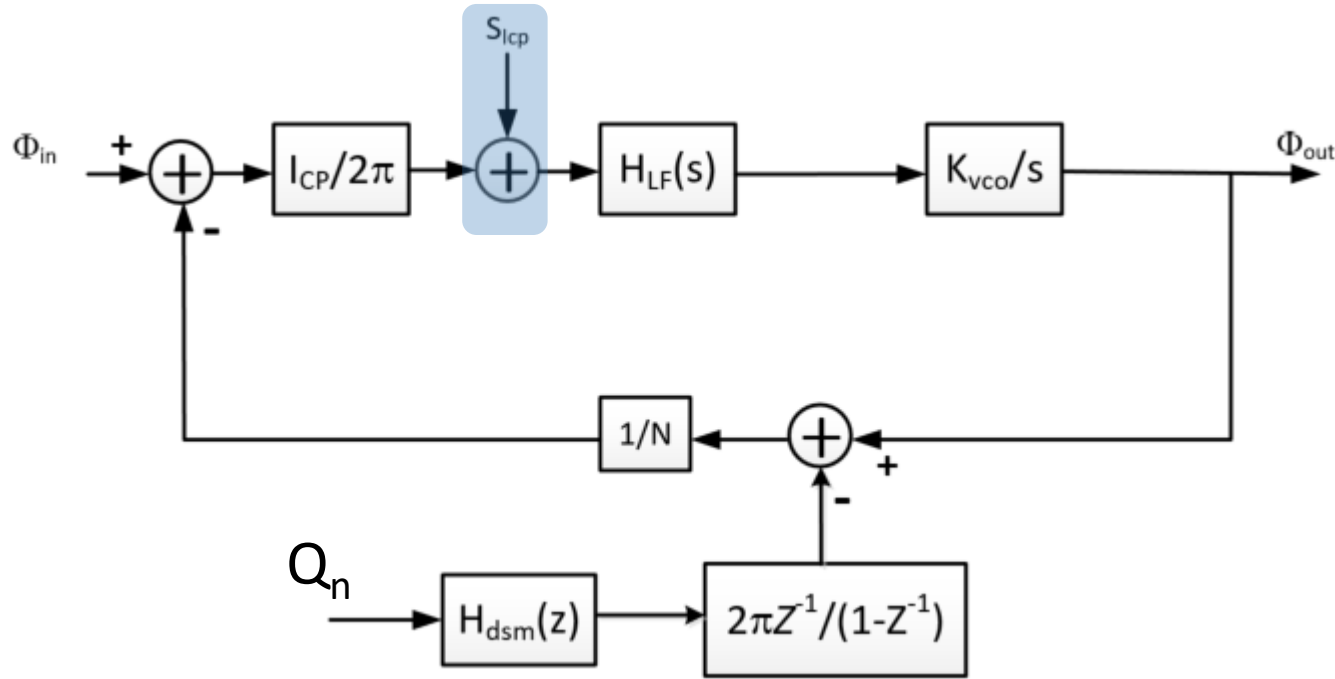
- Transfer function from reference input to PLL output:

$$H_{REF}(S) = \frac{N \cdot H_{OL}}{1 + H_{OL}} = \frac{\frac{I_{CP}}{2\pi} \cdot \frac{K_{VCO}}{C_Z + C_p} \left(1 + \frac{S}{\omega_Z}\right)}{\frac{S^3}{\omega_p} + S^2 + \frac{I_{CP}}{2\pi N} \cdot \frac{K_{VCO}}{C_Z + C_p} \left(1 + \frac{S}{\omega_Z}\right)}$$

- At low frequency, $H_{REF}(s) \approx N$. In-band reference noise is amplified by N times.
- At high frequency, reference noise is low pass filtered.
- Reference noise is low pass filtered by the loop. Reducing loop bandwidth will reduce reference noise at PLL output.
- Reference noise is dominant for in-band noise. It includes crystal oscillator noise, input buffer noise and frequency doubler or multiplier noise.
- When reference frequency is doubled, reference phase noise up by 6dB, N down by 6dB, overall phase noise from the reference at PLL output does not change.



PFD and Charge Pump Noise to PLL Output



- Feedforward path transfer function

$$H_{FF}(s) = \frac{K_{VCO}}{s} H_{LF}(S)$$

- Feedback path transfer function

$$H_{FB}(S) = \frac{I_{CP}}{2\pi N}$$

- Open loop transfer function

$$H_{OL}(s) = H_{FF} \cdot H_{FB} = \frac{I_{CP} K_{VCO}}{2\pi N s} H_{LF}(S)$$

- Transfer function from S_{CP} to Φ_{out} :

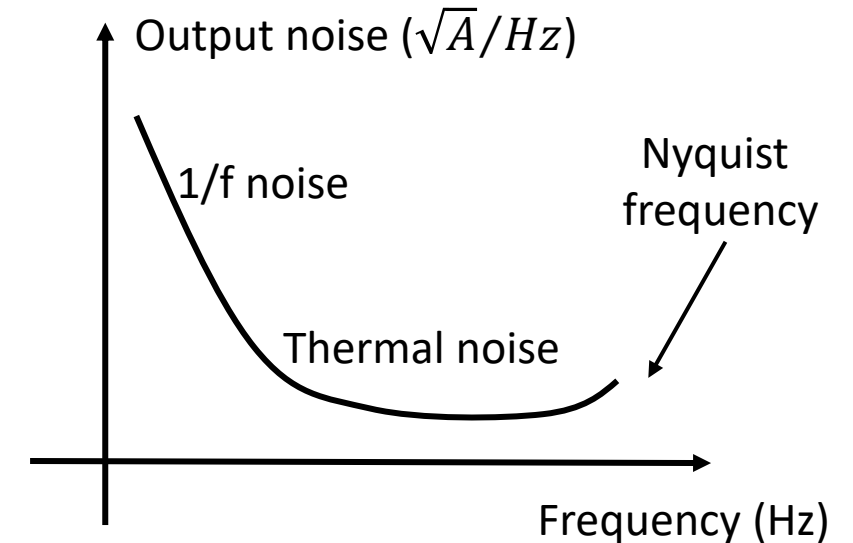
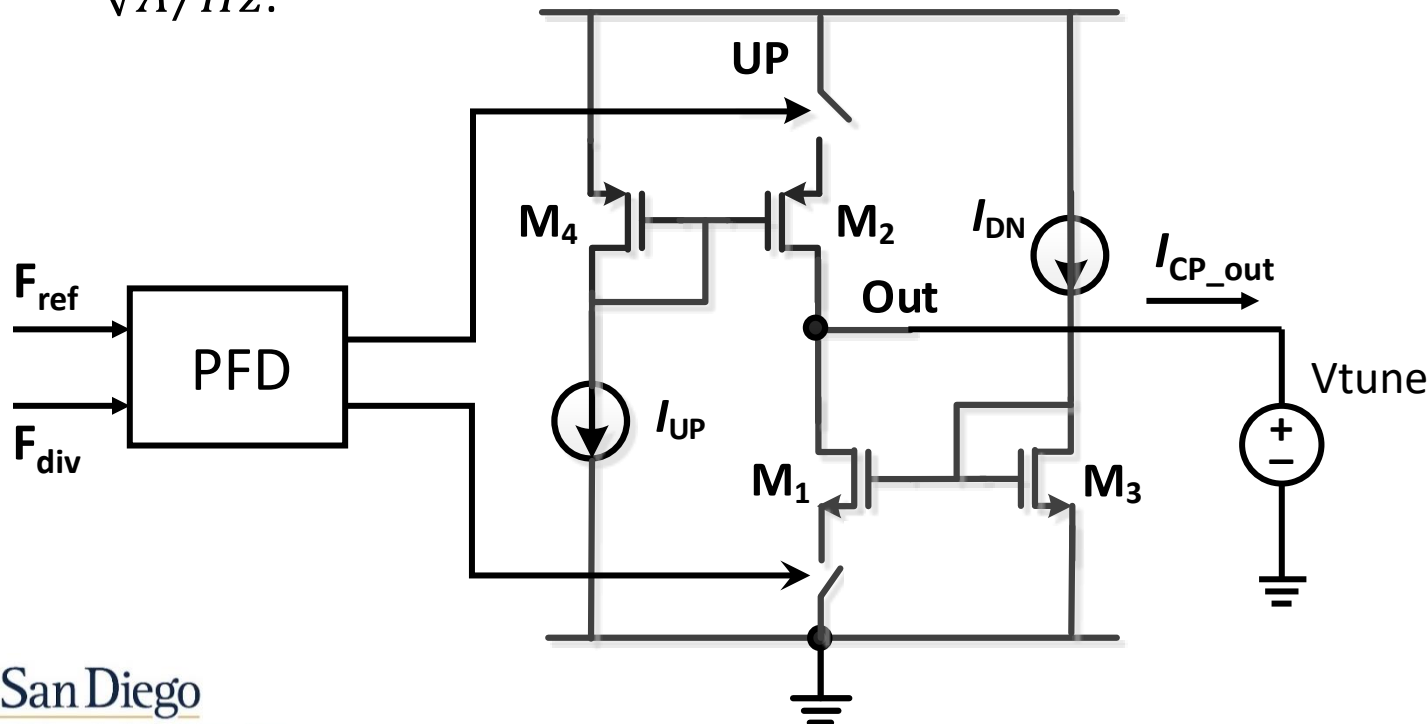
$$(S_{CP} - H_{FB} \Phi_{out}) H_{FF} = \Phi_{out}$$

$$H_{CP}(S) = \frac{\Phi_{out}}{S_{CP}} = \frac{H_{FF}}{1 + H_{OL}} = \frac{2\pi N}{I_{CP}} \cdot \frac{H_{OL}}{1 + H_{OL}}$$



PFD and Charge Pump Noise to PLL Output

- Charge pump noise simulation methodology
 - PFD and charge pump are simulated together
 - Provide F_{ref} and F_{div} input clock signals that are in phase – steady state charge pump noise when loop is locked, UP and DN pulses have minimum pulse width
 - Charge pump output is connected to a voltage source with DC voltage set to V_{tune}
 - Use PSS/Pnoise to simulate PFD/CP noise. PSS period is the reference period. Pnoise output in $\sqrt{A}/Hz$.

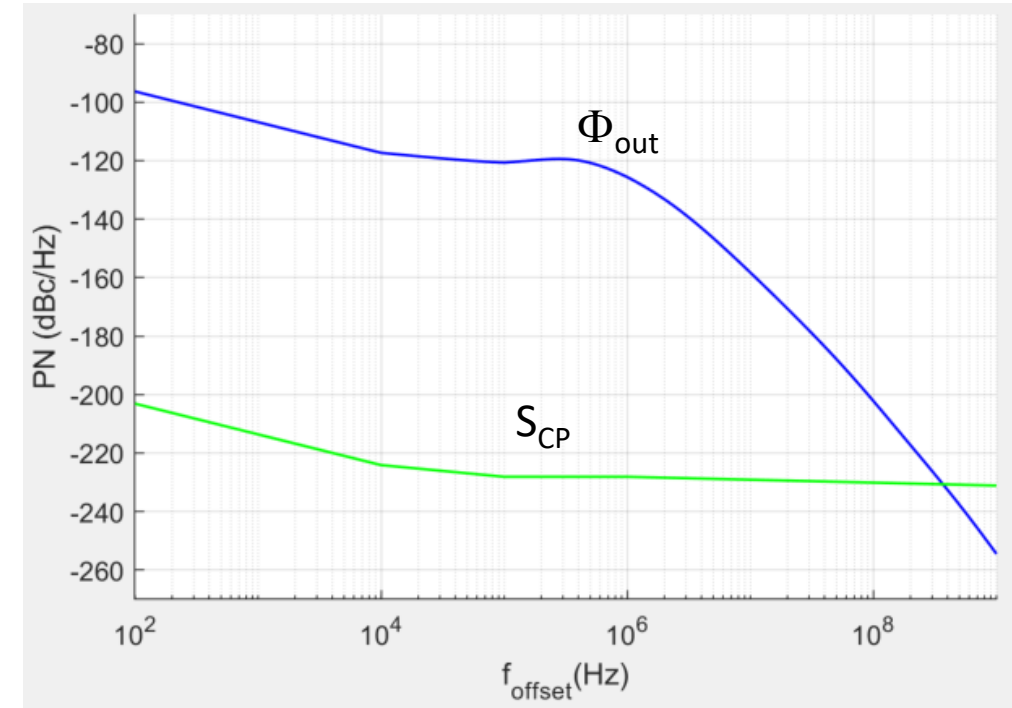


PFD and Charge Pump Noise to PLL Output

- Transfer function from PFD/CP to PLL output:

$$H_{CP}(S) = \frac{2\pi N}{I_{CP}} \cdot \frac{H_{OL}}{1+H_{OL}} = \frac{\frac{K_{VCO}}{C_Z+C_p} \left(1 + \frac{S}{\omega_Z}\right)}{\frac{S^3}{\omega_p} + S^2 + \frac{I_{CP}}{2\pi N} \cdot \frac{K_{VCO}}{C_Z+C_p} \left(1 + \frac{S}{\omega_Z}\right)}$$

- At low frequency, $H_{CP}(S) \approx \frac{2\pi N}{I_{CP}}$. In-band charge pump noise is amplified by N times.
- At high frequency, reference noise is low pass filtered.
- Charge pump noise is proportional to Up/Down pulse turn on time $\rightarrow$ tradeoff between charge pump linearity and phase noise.



Charge Pump Noise Performance (from Lecture 4)

- Transfer function from charge pump output to

PLL output $|H_{CP}(s)|^2 \propto \left(\frac{2\pi}{I_{CP}}\right)^2$

- Current source thermal noise

$$S_{I_{CP},th} = \overline{I_{n,th}^2} = 4kT\gamma g_m \propto \sqrt{I_{CP}}$$

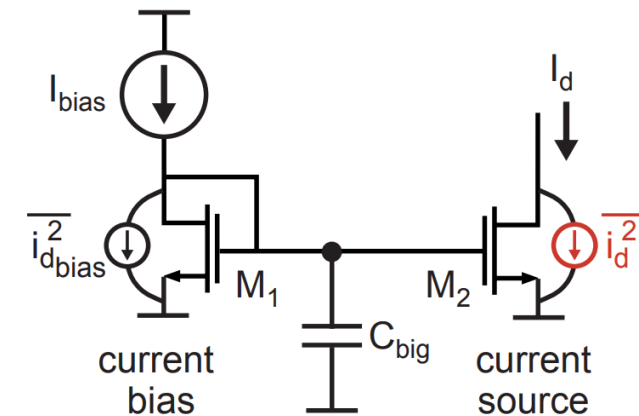
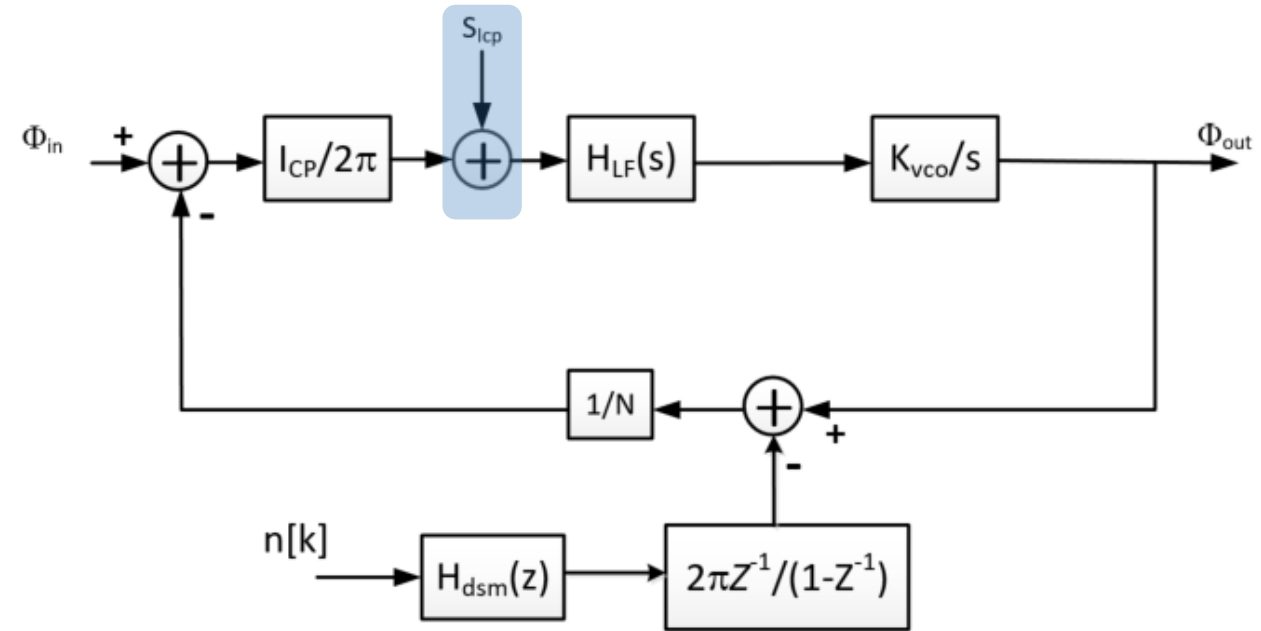
- Current source flicker noise

$$S_{I_{CP},1/f} = \overline{I_{n,1/f}^2} \frac{K}{C_{ox}WL} \cdot \frac{1}{f} \cdot g_m^2 \propto I_{CP}$$

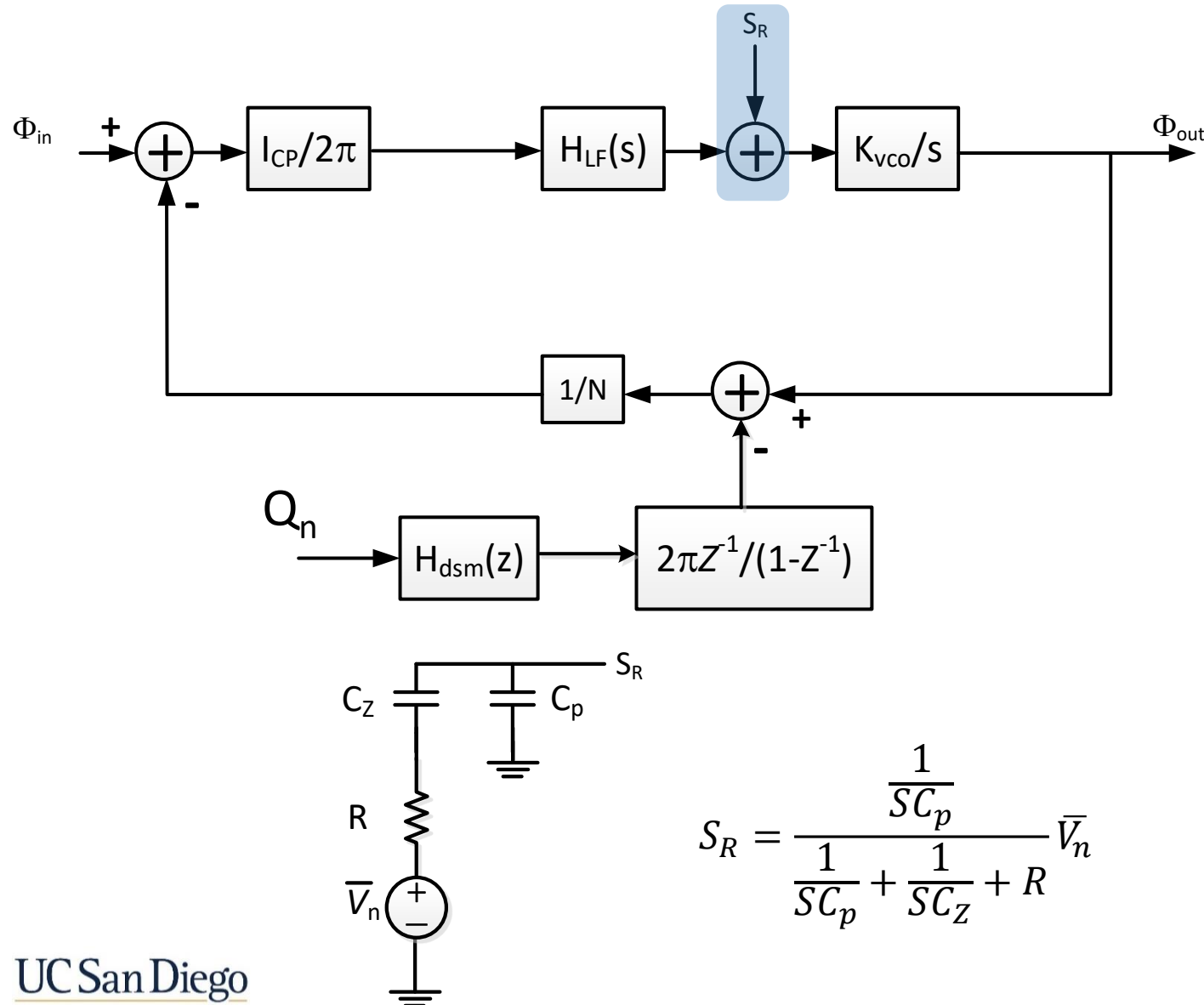
- PLL output noise from charge pump $S_{\Phi_{out}}(f) =$

$$(S_{I_{CP},th} + S_{I_{CP},1/f}) |H_{CP}(f)|^2 \propto \frac{a\sqrt{I_{CP}} + bI_{CP}}{I_{CP}^2}$$

- Increasing charge pump current increases signal to noise ratio $\rightarrow$ tradeoff between power consumption and phase noise.



Loop Filter Resistor Noise to PLL Output



- Feedforward path transfer function

$$H_{FF}(s) = \frac{K_{VCO}}{s}$$

- Feedback path transfer function

$$H_{FB}(S) = \frac{I_{CP}}{2\pi N} H_{LF}(S)$$

- Open loop transfer function

$$H_{OL}(s) = H_{FF} \cdot H_{FB} = \frac{I_{CP} K_{VCO}}{2\pi N s} H_{LF}(S)$$

- Transfer function from \$S_R\$ to \$\Phi_{out}\$:

$$(S_R - H_{FB} \Phi_{out}) H_{FF} = \Phi_{out}$$

$$H_R(S) = \frac{\Phi_{out}}{S_R} = \frac{H_{FF}}{1 + H_{OL}} = \frac{\frac{K_{VCO}}{S}}{1 + H_{OL}}$$

$$H_{V_{Rn}}(S) = \frac{\Phi_{out}}{\bar{V}_n} = \frac{\frac{K_{VCO}}{S}}{1 + H_{OL}} \cdot \frac{\frac{1}{SC_p}}{\frac{1}{SC_p} + \frac{1}{SC_Z} + R}$$

$$S_R = \frac{\frac{1}{SC_p}}{\frac{1}{SC_p} + \frac{1}{SC_Z} + R} \bar{V}_n$$

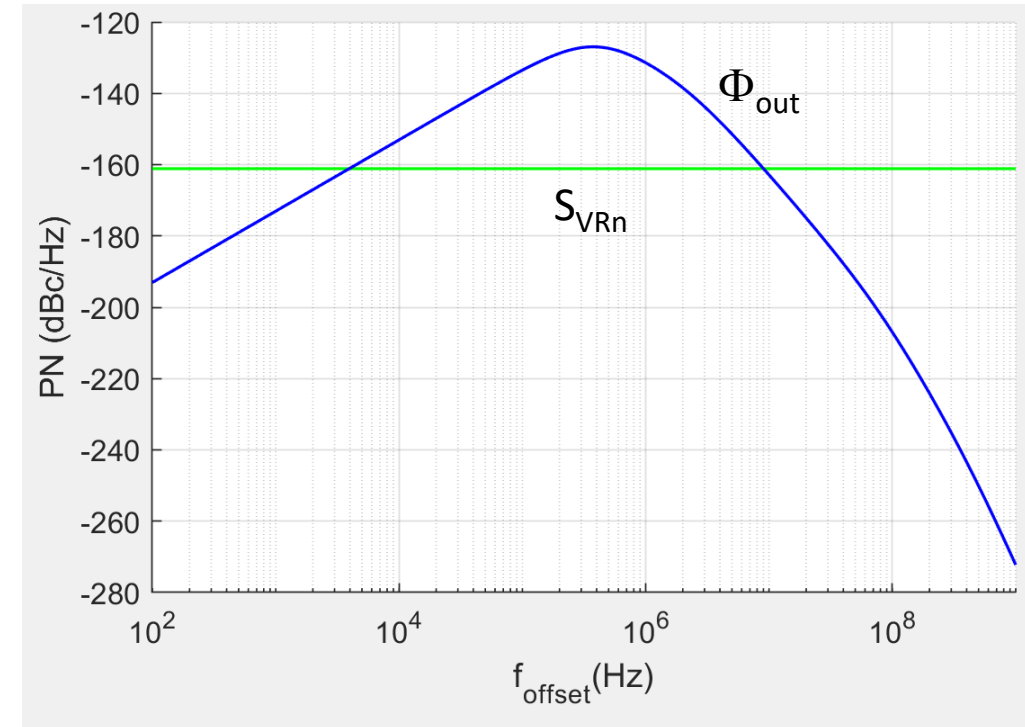
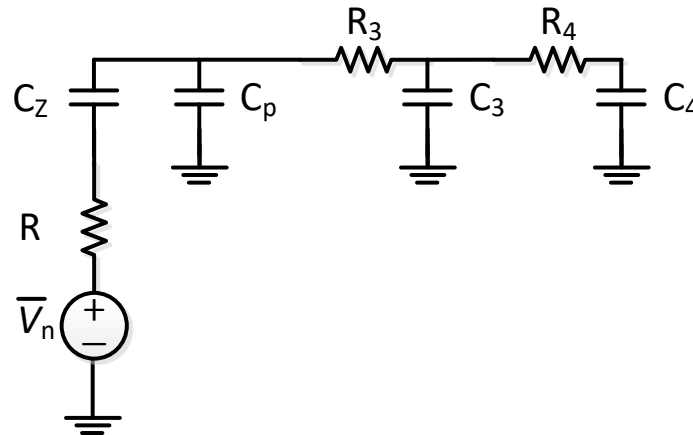


Loop Filter Resistor Noise to PLL Output

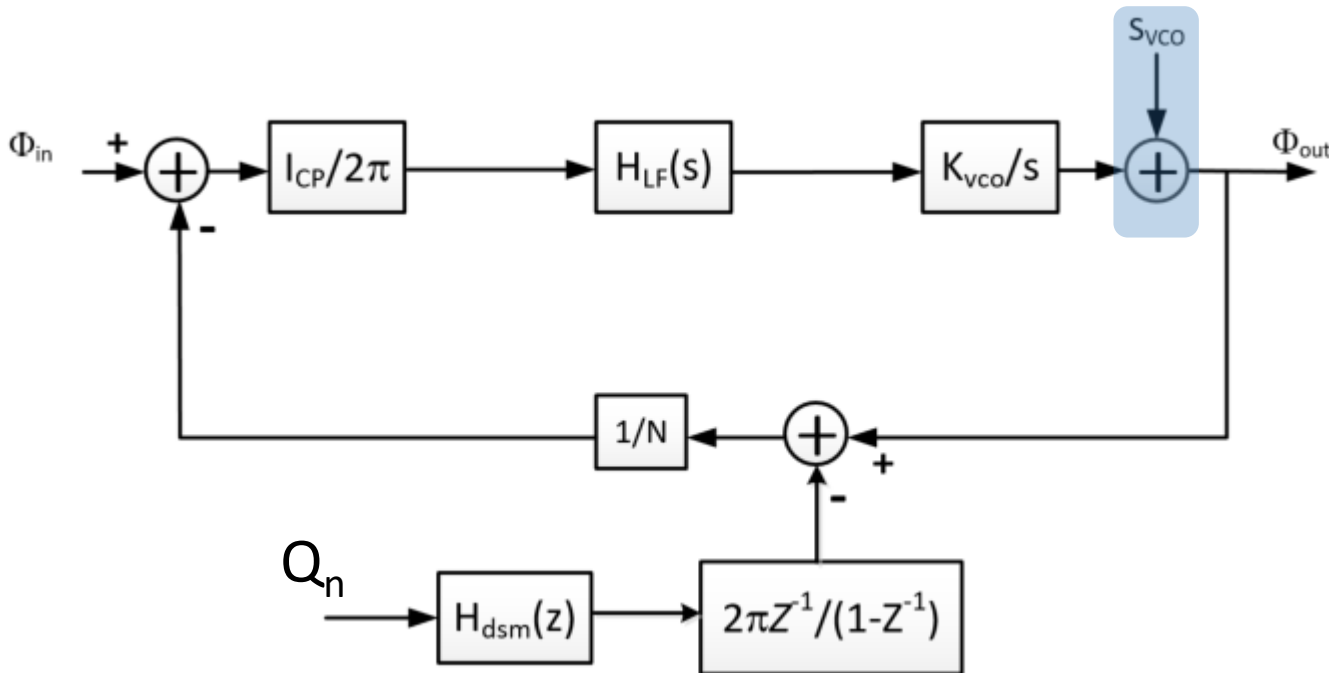
- Transfer function from loop filter resistor to PLL output:

$$H_{V_{Rn}}(S) = \frac{\frac{K_{VCO}}{S}}{1 + H_{OL}} \cdot \frac{\frac{1}{SC_p}}{\frac{1}{SC_p} + \frac{1}{SC_Z} + R}$$

- At low frequency, resistor noise is attenuated by loop capacitor.
- At high frequency, resistor noise is low pass filtered.
- Higher order loop filter resistors also contribute noise
- Usually loop filter noise is not a significant part of total phase noise.



VCO Phase Noise to PLL Output



- Feedforward path transfer function

$$H_{FF}(s) = 1$$

- Feedback path transfer function

$$H_{FB}(S) = \frac{I_{CP}K_{VCO}}{2\pi NS} H_{LF}(S)$$

- Open loop transfer function

$$H_{OL}(s) = H_{FF} \cdot H_{FB} = \frac{I_{CP}K_{VCO}}{2\pi NS} H_{LF}(S)$$

- Transfer function from S_{VCO} to Φ_{out} :

$$(S_{VCO} - H_{FB}\Phi_{out})H_{FF} = \Phi_{out}$$

$$H_{VCO}(S) = \frac{\Phi_{out}}{S_{VCO}} = \frac{H_{FF}}{1 + H_{OL}} = \frac{1}{1 + H_{OL}}$$

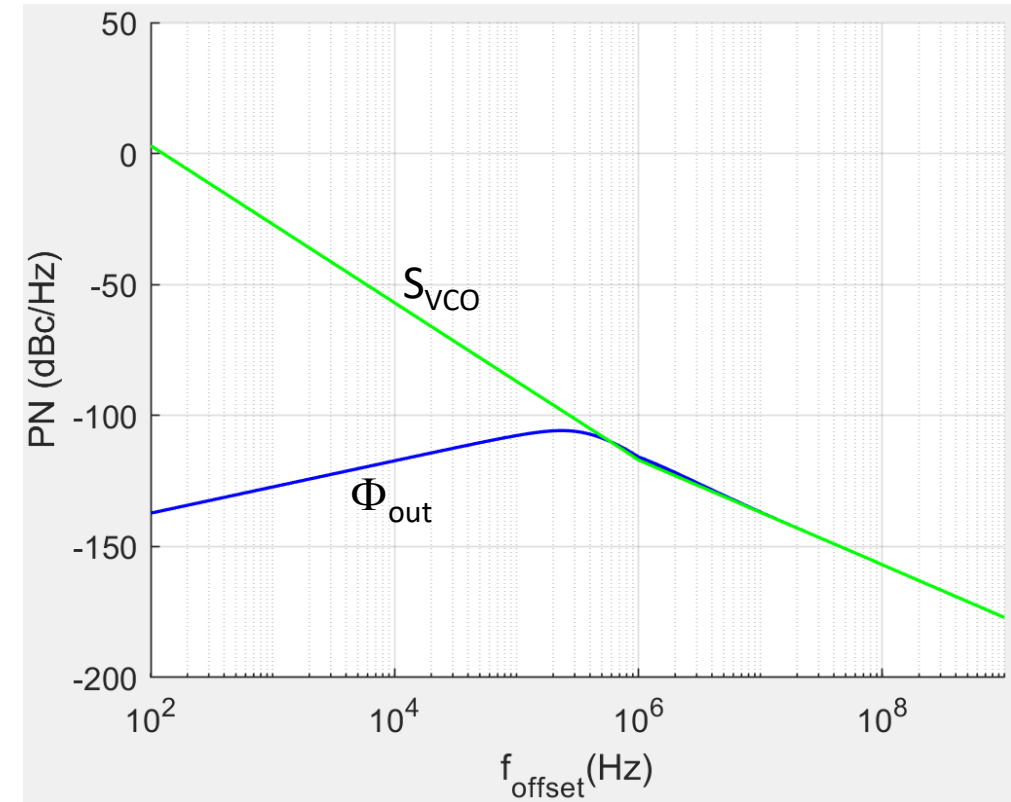


VCO Phase Noise to PLL Output

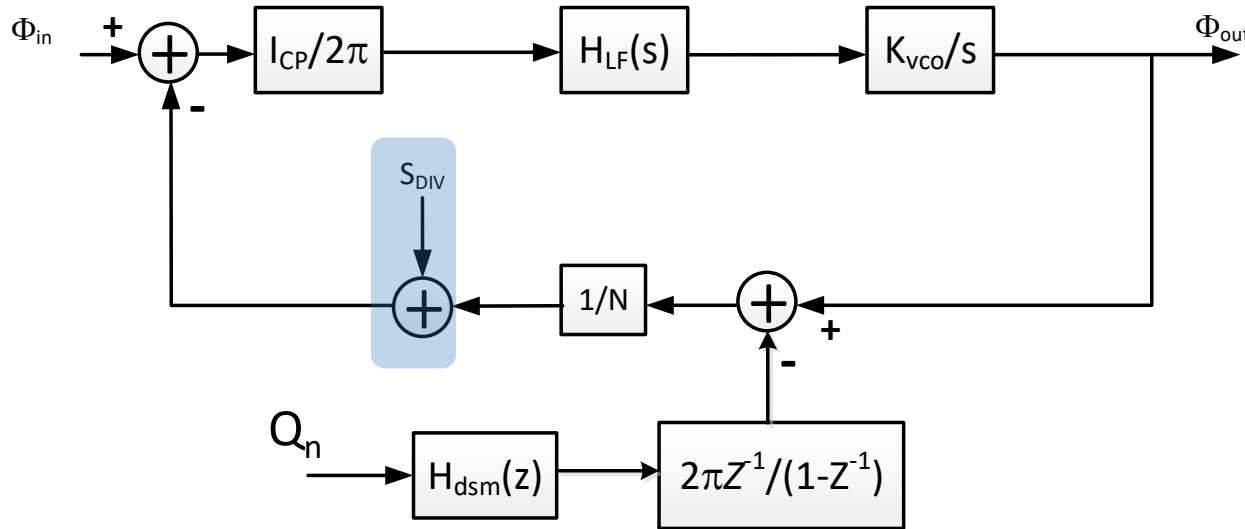
- Transfer function from VCO to PLL output:

$$H_{VCO}(S) = \frac{1}{1+H_{OL}} = \frac{\frac{S^3}{\omega_p} + S^2}{\frac{S^3}{\omega_p} + S^2 + \frac{I_{CP}}{2\pi N} \cdot \frac{K_{VCO}}{C_Z + C_P} \left(1 + \frac{S}{\omega_Z}\right)}$$

- At low frequency, $H_{VCO}(S) \approx \frac{2\pi N(C_Z + C_P)S^2}{I_{CP}K_{VCO}}$, VCO phase noise is attenuated.
- At high frequency, $H_{VCO}(S) \approx 1$, PLL output follows VCO noise profile.
- VCO phase noise is high pass filtered by the loop. Increasing loop bandwidth reduces VCO noise at output.
- Tradeoff between reference path noise and VCO noise in choosing optimal loop bandwidth.
- Use PSS/Pnoise to simulate VCO phase noise.



N Divider Noise to PLL Output



- Feedforward path transfer function

$$H_{FF}(s) = \frac{I_{CP}K_{VCO}}{2\pi s} H_{LF}(s)$$

- Feedback path transfer function

$$H_{FB}(s) = \frac{1}{N}$$

- Open loop transfer function

$$H_{OL}(s) = H_{FF} \cdot H_{FB} = \frac{I_{CP}K_{VCO}}{2\pi N s} H_{LF}(s)$$

- Transfer function from S_{DIV} to Φ_{out} :

$$(S_{DIV} - H_{FB}\Phi_{out})H_{FF} = \Phi_{out}$$

$$H_{DIV}(s) = \frac{\Phi_{out}}{S_{DIV}} = \frac{H_{FF}}{1 + H_{OL}} = \frac{N \cdot H_{OL}}{1 + H_{OL}}$$

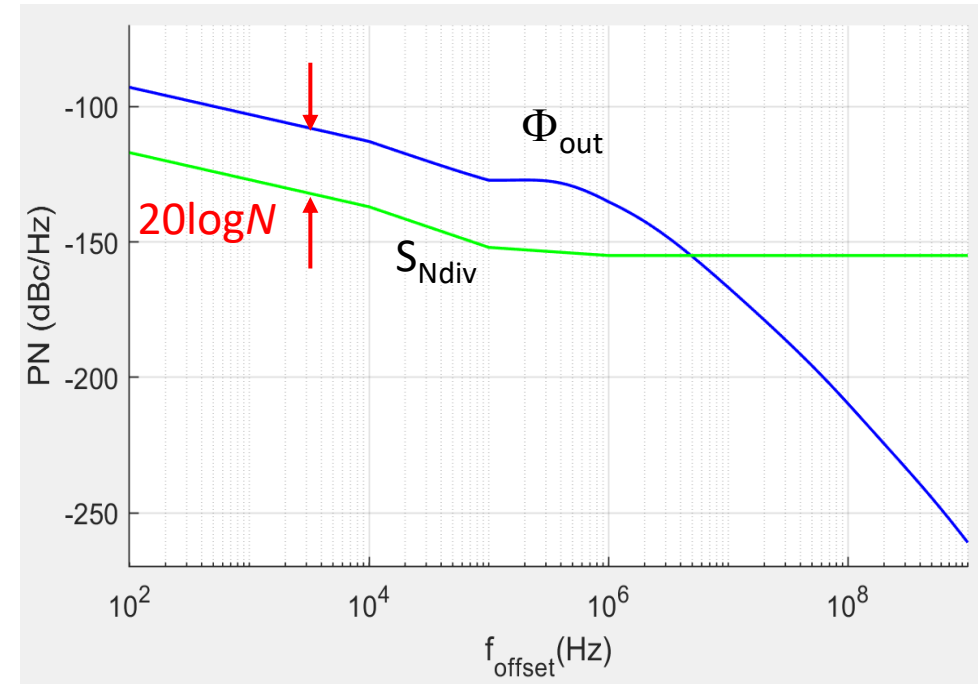


N Divider Noise to PLL Output

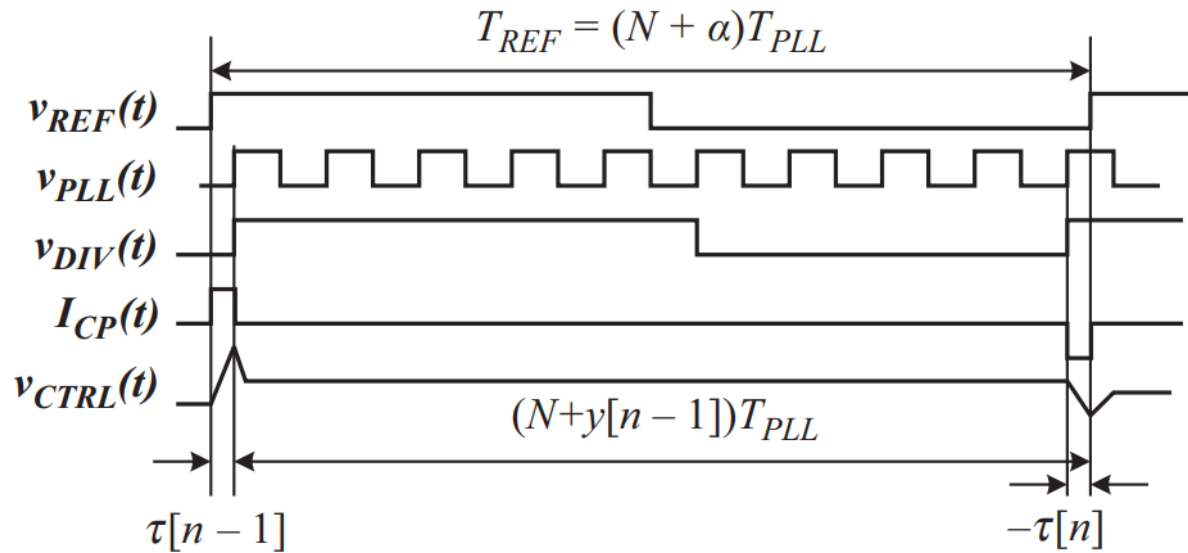
- Transfer function from N divider to PLL output:

$$H_{DIV}(S) = \frac{N \cdot H_{OL}}{1 + H_{OL}} = \frac{\frac{I_{CP}}{2\pi} \cdot \frac{K_{VCO}}{C_Z + C_p} \left(1 + \frac{S}{\omega_Z}\right)}{\frac{S^3}{\omega_p} + S^2 + \frac{I_{CP}}{2\pi N} \cdot \frac{K_{VCO}}{C_Z + C_p} \left(1 + \frac{S}{\omega_Z}\right)}$$

- At low frequency, $H_{DIV}(s) \approx N$. In-band reference noise is amplified by N times.
- At high frequency, reference noise is low pass filtered.
- N divider noise transfer function is identical to reference noise transfer function. Divider noise is low pass filtered by the loop. Reducing loop bandwidth will reduce divider noise at PLL output.
- Divider noise can be high depending on implementation and current consumption. Usually use prescaler output to retime divider output to reduce noise.



DSM Model in PLL



1. $\tau[n] = \tau[n - 1] + (y[n - 1] - \alpha)T_{PLL}$
2. $Q_n[n - 1] = y[n - 1] - \alpha$
3. $\phi[n] = 2\pi \frac{\tau[n]}{T_{ref}} = \phi[n - 1] + (y[n - 1] - \alpha) \frac{2\pi \cdot T_{PLL}}{T_{ref}}$
4. $\phi[n] = \phi[n - 1] + (y[n - 1] - \alpha) \frac{2\pi}{N + \alpha}$

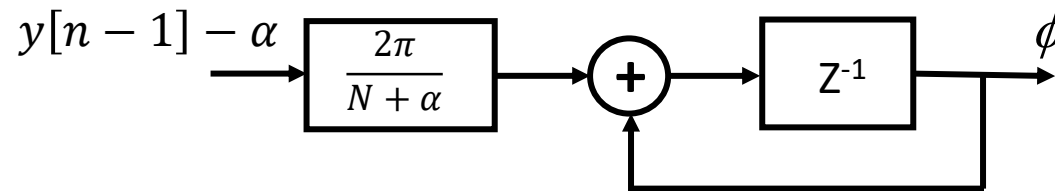
- Ndiv divide ratio = $N + \alpha$, $f_{PLL} = (N + \alpha)f_{ref}$ and $T_{PLL} = T_{ref}/(N + \alpha)$
- Instantaneous divide ratio = $N + y[n]$, $avg(y[n]) = \alpha$
- Difference b/w the instantaneous divide ratio and the average divide ratio is quantization error: $y[n - 1] - \alpha$
- Even in steady state, $f_{div}(t) = f_{ref}$ only in an average sense
- In each reference period, $f_{div}(t)$ and f_{ref} exhibit instantaneous phase error $\phi[n]$ – caused by quantization error
- From Eqn 4), $\phi[n]$ is an accumulation of quantization error with a scaling factor of $2\pi/(N + \alpha)$



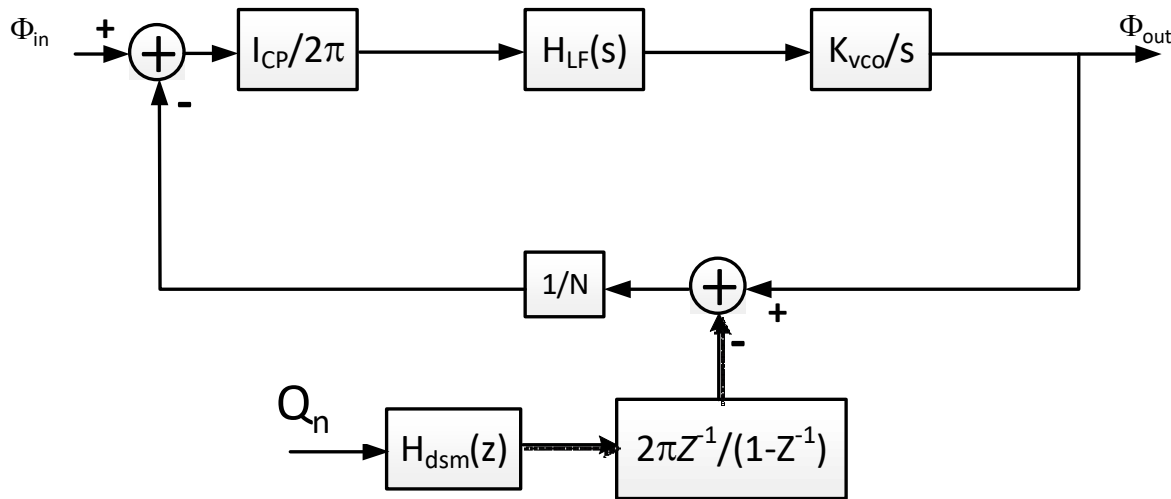
DSM Model in PLL

$$\phi[n] = \phi[n - 1] + (y[n - 1] - \alpha) \frac{2\pi}{N + \alpha}$$

- Quantization error $Q_n[n - 1] = y[n - 1] - \alpha$
- $\phi[n]$ is an accumulation of quantization error Q_n with a scaling factor of $2\pi/(N+\alpha)$
- In Z domain, $\Phi(Z) = \frac{2\pi}{N+\alpha} \cdot \frac{Z^{-1}}{1-Z^{-1}} Q_n(Z)$
- Quantization error is the Ndiv divide ratio error, which is proportional to frequency error
- Phase is the integration/accumulation of frequency $\rightarrow \frac{Z^{-1}}{1-Z^{-1}}$ factor



DSM Quantization Noise to PLL Output



- Feedforward path transfer function

$$H_{FF}(s) = \frac{I_{CP}K_{VCO}}{2\pi NS} H_{LF}(S)$$

- Feedback path transfer function

$$H_{FB}(S) = 1$$

- Open loop transfer function

$$H_{OL}(s) = H_{FF} \cdot H_{FB} = \frac{I_{CP}K_{VCO}}{2\pi NS} H_{LF}(S)$$

- Transfer function from $S_{\Phi_{dsm}}$ to Φ_{out} :

$$(S_{\Phi_{dsm}} - H_{FB}\Phi_{out})H_{FF} = \Phi_{out}$$

$$H_{\Phi_{dsm}}(S) = \frac{\Phi_{out}}{S_{\Phi_{dsm}}} = \frac{H_{FF}}{1 + H_{OL}} = \frac{H_{OL}}{1 + H_{OL}}$$

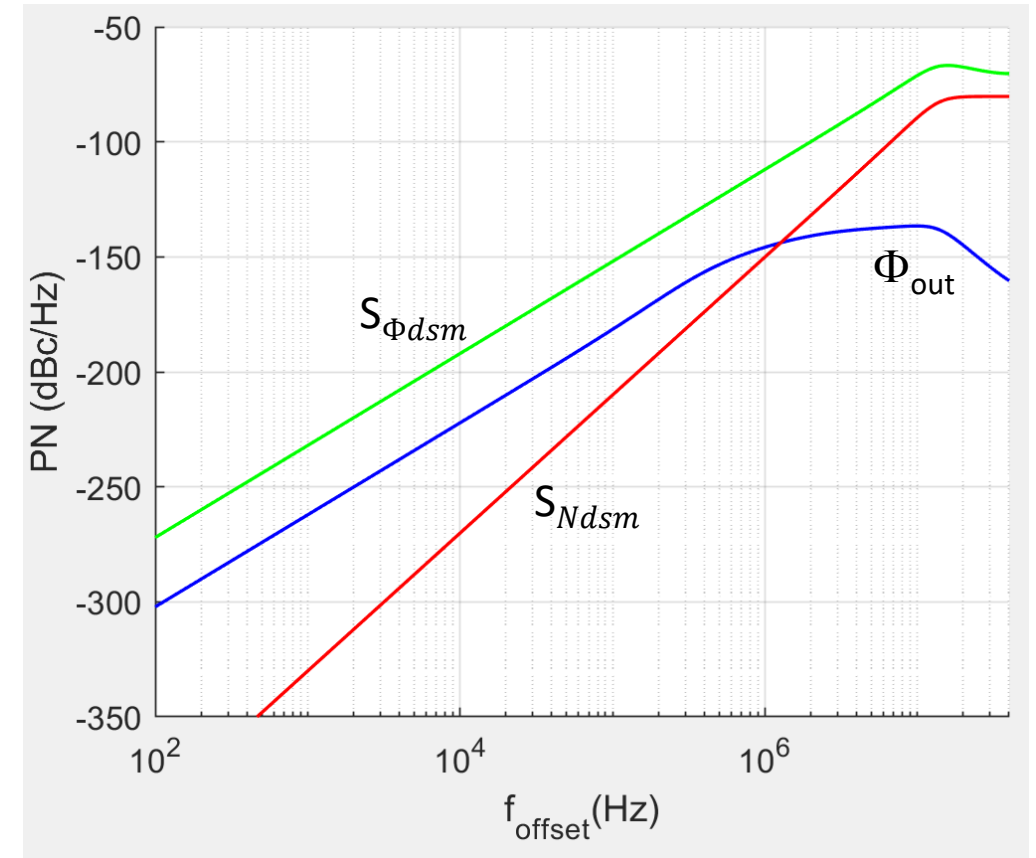
- Transfer function from Q_n to Φ_{out} also includes DSM noise transfer function and integrator.

$$H_{Qn}(S) = \frac{\Phi_{out}}{S_{Qn}} = \frac{H_{OL}}{1 + H_{OL}} \cdot H_{dsm}(Z) \cdot 2\pi Z^{-1} \frac{1}{1 - Z^{-1}} \Big|_{Z=e^{ST}}$$

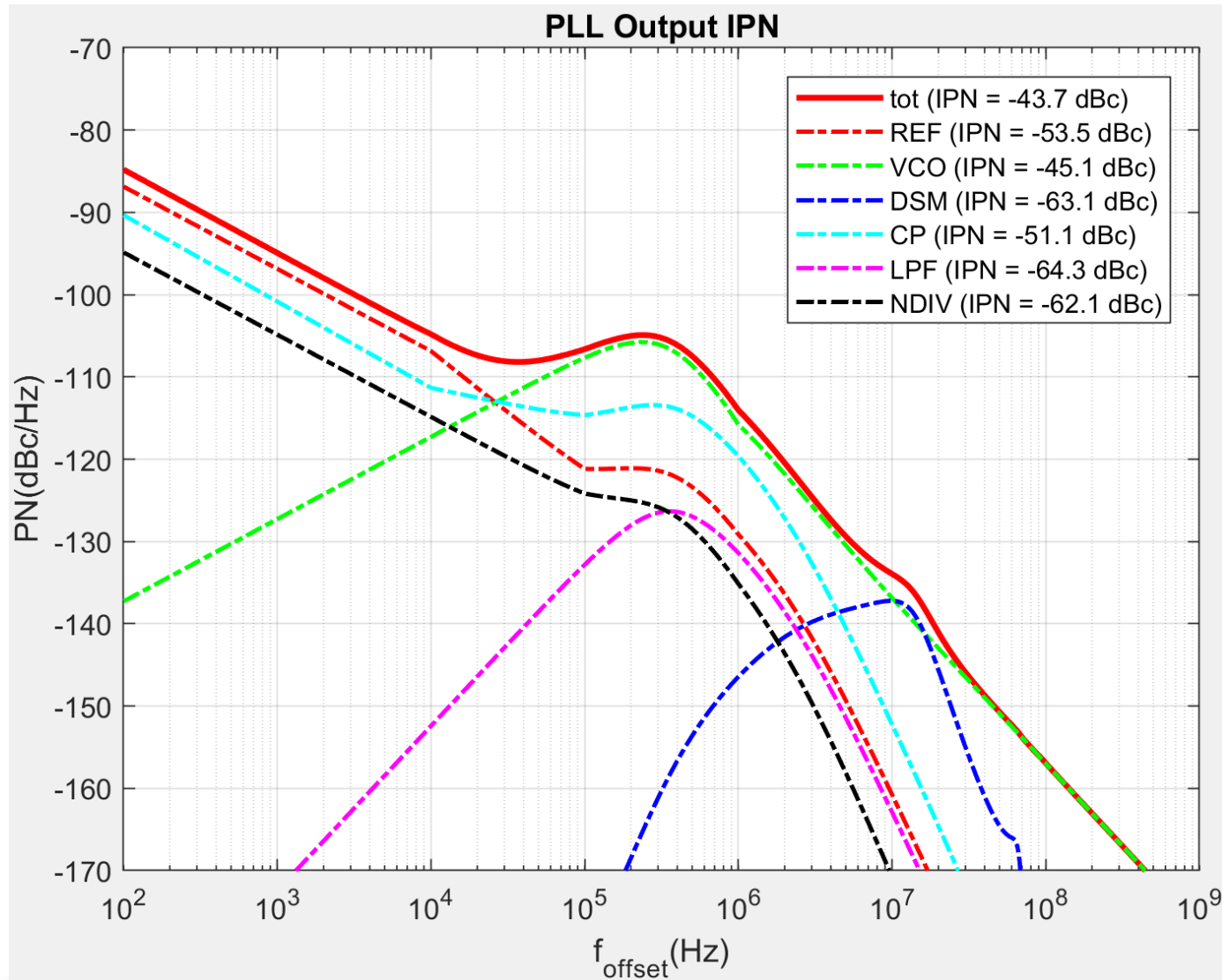


DSM Quantization Noise to PLL Output

- Integration to convert divide ratio N quantization noise to phase quantization noise reduces DSM order by one.
- Loop filter order needs to be at least one order higher than DSM noise for out of band noise attenuation.
- DSM noise dominates band edge spot phase noise
- Increase DSM clock frequency (F_v)
 - Reduces in-band noise
 - Pushes band edge noise peak to higher frequency that can be more effectively attenuated by loop filter.



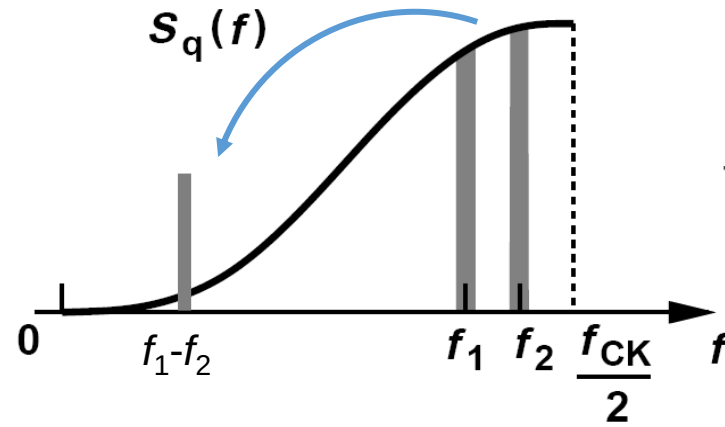
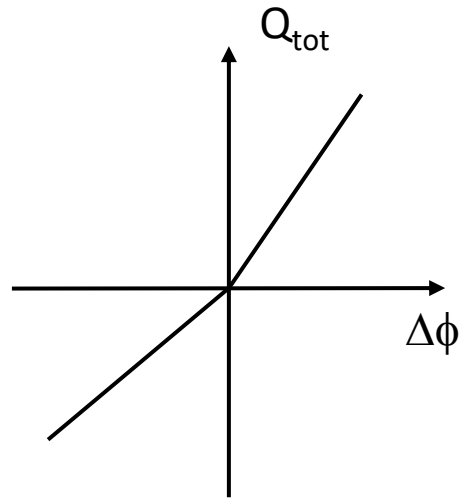
Composite PLL Output Phase Noise and IPN



- Reference signal noise is low pass filtered and dominates low offset frequency phase noise
- VCO noise is high pass filtered and dominates high offset frequency phase noise starting around 3dB bandwidth
- PFD, CP and N divider are low pass filtered
- DSM quantization noise is high pass shaped and low pass filtered
- Optimal loop bandwidth for phase noise is set to the value at which reference path and VCO contributes approximately equal phase noise at PLL output.



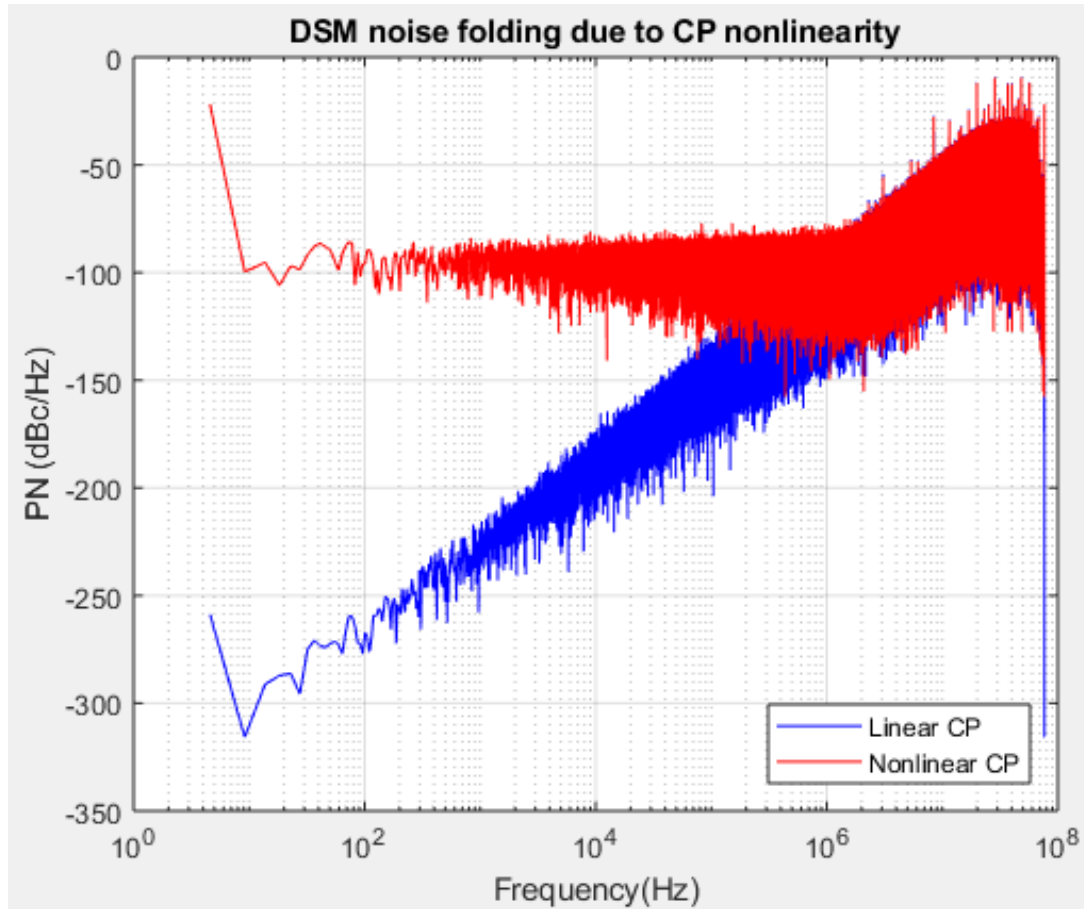
DSM Noise Folding



- Charge pump nonlinearity due to Up and Dn current mismatch
- $Q_{\text{tot}} \approx I_{\text{avg}}\Delta T_{\text{in}} + \alpha\Delta T_{\text{in}}^2 - b$
- The nonlinear term $\alpha\Delta T_{\text{in}}^2$ generates 2nd order product at $\Delta f = f_1 - f_2$ that folds out of band noise into PLL bandwidth. Once noise is folded to in-band, loop filter cannot attenuate it.



DSM Noise Folding



- DSM quantization noise folded to in-band due to nonlinearity in the loop, including charge pump and divider in analog PLLs.
- Noise folding is more significant in high order DSM.
- Charge pump nonlinearity can be improved by
 - Improving Up and Dn current matching
 - Adding a leaker current to move charge pump operating region to one side.



PLL Loop Bandwidth Optimization

- PLL loop bandwidth considerations
 - Phase noise
 - Wider loop bandwidth attenuates VCO in-band phase noise
 - Narrower loop bandwidth attenuates reference path out of band phase noise and DSM quantization noise
 - Lock time
 - Wider loop bandwidth allows faster settling and shorter lock time
 - Spurs
 - Narrower bandwidth has more attenuation on the reference spurs and allows less in-band fractional spurs
 - Loop filter integration
 - Wider bandwidth requires smaller loop filter capacitance and reduces integration cost
 - Loop bandwidth $< 10\%$ of the reference frequency for stability considerations



Matlab Code for IPN Calculation

```
% Analog CP-PLL IPN Calculation
close all;
clear all;
%Parameter DEFINITIONS
uA = 1e-6;
fF = 1e-15;
pF = 1e-12;
pS = 1e-12;
MHz = 1e6;
k = 1.38e-23;
T = 273.15 + 55;

Fref = 160 * MHz;
Fvco = 2570 * 2 * MHz;
lo_div = 1;
BW3db = 1e6;
kpole2 = 4;
Kvco = 2 * pi * 30 * MHz;

%CP Noise
f_cp = [1e3 1e4 1e5 1e6 10e6];
Lcp_dB_scp = [-208.5 -219 -223 -223 -224];
%XO
f_xo = [1e3 1e4 1e5 1e6 10e6];
Lref_dB = [-127 -137 -152 -155 -155];
%XO_Gen
f_xo_gen = [1e3 1e4 1e5 1e6 10e6];
Lrefgen_dB = [-123 -138 -158 -161 -161];
%VCO Noise
f_vco = [1e4 1e5 2e5 4e5 8e5 1e6 6e6 10e6 20e6 45e6
80e6 100e6 190e6];
Lvco_dB = [-57 -87 -96 -105 -114 -117 -132.5 -137 -143
-150 -155 -157 -162.6];
%Pre & NDIV Noise
f_pre = [1e4 1e5 1e6];
L_pre = [-132 -152 -155];

Icp = 465e-6;
R1 = 8.5e3;
C1 = 100 * pF;
C2 = 5.7e-12;
R3 = 2e3;
C3 = 0.45e-12;
R4 = 2e3;
C4 = 0.45e-12;
%Integration BW
IPN_Fstart = 1e3;
IPN_Fstop = 50e6;
Ndiv = Fvco / Fref;
Tref = 1 / Fref;
fstart = 100;
fstop = 1e9;
num_pt_per_dec = 1000;
f = logspace(log10(fstart), log10(fstop), num_pt_per_dec);
%__LAPACE AND Z VARIABLES
z = tf('z', Tref);
s = tf([1 0], 1, 'variable', 's');
Z = exp(j * 2 * pi * f * Tref);
S = j * 2 * pi * f;
% CP Gain
Kcp = Icp / (2 * pi);
%fl = 1 / (R1 * C2 * 2 * pi);
Z1 = 1 / (s * C3) * (R4 + 1 / (s * C4)) / (1 / (s * C3) + 1 / (s * C4) + R4);
Zi = 1 / (s * C2) * (R1 + 1 / (s * C1)) / (1 / (s * C2) + 1 / (s * C1) + R1);
F = Zi * (R3 + Z1) / (Zi + R3 + Z1) * Z1 / (R3 + Z1);

As = Kcp * F * Kvco / Ndiv / s;
GHs = freqs(As.num{:,}, As.den{:,}, 2 * pi * f);
Ts = Ndiv * GHs / (1 + GHs);

Hvco = 1 ./ (1 + GHs);
Hdsm = GHs ./ (1 + GHs);

Fstart = IPN_Fstart;
Fstop = IPN_Fstop;
IPN_Range = find(f >= Fstart & f <= Fstop);
%Ref Clock SSB
Lref_tot_dB = interp1(10 * log10(f_xo), Lref_dB, 10 * log10(f),
'linear', 'extrap');
So_ref_dB = Lref_tot_dB + 20 * log10(abs(squeeze(Ts))) +
20 * log10(1 / lo_div);
IPN =
2 * trapz(f(IPN_Range), 10.^((So_ref_dB(IPN_Range)) / 10));
IPN_ref = IPN;
%Ref Buf SSB
Lrefgen_tot_dB = interp1(10 * log10(f_xo_gen), Lrefgen_dB,
10 * log10(f), 'linear', 'extrap');
So_refgen_dB = Lrefgen_tot_dB + 20 * log10(abs(squeeze(Ts)))
+ 20 * log10(1 / lo_div);
IPN =
2 * trapz(f(IPN_Range), 10.^((So_refgen_dB(IPN_Range)) / 10));
IPN_refgen = IPN;
%VCO SSB
L =
interp1(10 * log10(f_vco), Lvco_dB, 10 * log10(f), 'linear', 'extra
p');
So_vco_dB = L + 20 * log10(abs(squeeze(Hvco))) +
20 * log10(1 / lo_div);
IPN =
2 * trapz(f(IPN_Range), 10.^((So_vco_dB(IPN_Range)) / 10));
IPN_vco = IPN;
```



Matlab Code for IPN Calculation

`%Delta Sigma Modulator SSB`

```
b1 = -0.972;
b2 = 0.579;
b3 = -0.107;

Sf_dsm = (1/(12*Fref)).*abs(((1-
Z.^1).^3)./(1+b1.*Z.^1+b2.*Z.^2+b3.*Z.^3)).^2;
L_dsm = Sf_dsm.*abs(Z.^-1*(2*pi)./(1-Z.^-1)).^2;
L_dsm_dB = 10*log10(L_dsm);

So_dsm = L_dsm.*abs(Hdsm).^2;
So_dsm_dB = 10*log10(So_dsm)-20*log10(Ndiv);
IPN =
2*trapz(f(IPN_Range),10.^((So_dsm_dB(IPN_Range))/10));
IPN_dsm = IPN;

%CP Noise SSB
Lcp_dB_scp_scaled = Lcp_dB_scp +
10*log10(Icp/(1500*uA));
L_cp_scp = interp1(10*log10(f_cp),Lcp_dB_scp_scaled,
10*log10(f),'linear','extrap');

So_cp_dB = L_cp_scp + 20*log10(abs(squeeze(Ts/Kcp))) +
20*log10(1/lo_div);

IPN_cp =
2*trapz(f(IPN_Range),10.^((So_cp_dB(IPN_Range))/10));
Z_C1 = (1./(S*C1));
Z_C2 = (1./(S*C2));
Z_C3 = (1./(S*C3));
Z_C4 = (1./(S*C4));
```

```
Z1_lpf = R4 + Z_C4;
Z2_lpf = (Z_C3.*(R4+Z_C4))./(Z_C3+R4+Z_C4);
Z3_lpf = R3 + Z2_lpf;
Z4_lpf = (Z_C2.*Z3_lpf)./(Z_C2+Z3_lpf);
Z5_lpf = (Z_C2.*(R1+Z_C1))./(Z_C2+R1+Z_C1);
Z6_lpf = (Z_C3.*(R3+Z5_lpf))./(Z_C3+R3+Z5_lpf);

GR1 =
(Z4_lpf./(Z4_lpf+R1+Z_C1)).*(Z2_lpf./(Z2_lpf+R3)).*(Z_C
4./(Z_C4 +R4));
GR2 = (Z_C4./(Z_C4+R4)).*(Z2_lpf./(Z2_lpf+R3+Z5_lpf));
GR3 = (Z_C4./(Z_C4+R4+Z6_lpf));

L_lpf =
2*k*T*(R1*(abs(GR1).^2)+0*(abs(GR2).^2)+0*(abs(GR3).^2)
);
L_lpf_dB = 10*log10(L_lpf);
So_lpf = L_lpf.*(abs((Kvco./S).*Hvco).^2);
So_lpf_dB = 10*log10(So_lpf)+20*log10(1/lo_div);
f_lpf = [1e3 10e3 50e3 100e3 500e3 1e6 1.5e6 1.6e6
1.7e6 10e6];
IPN =
2*trapz(f(IPN_Range),10.^((So_lpf_dB(IPN_Range))/10));
IPN_lpf = IPN;
```

`%IPN Plot`

```
So_tot = 10.^(So_ref_dB./10) + 10.^(So_refgen_dB./10) +
10.^(So_vco_dB./10) + 10.^(So_dsm_dB./10) +
10.^(So_cp_dB./10) + 10.^(So_lpf_dB./10);
So_tot_dB = 10*log10(So_tot);
IPN = 2*trapz(f(IPN_Range),10.^(So_tot_dB(IPN_Range)./10));
IPN_tot = IPN;

fig_size = [200,200,910,550];
h_ipn=figure('position',fig_size);
semilogx(f,So_tot_dB,'r','LineWidth',3); grid on;
xlabel('foffset/Hz');
ylabel('PN/dBc/Hz');

axis([100 10^9 -180 -70]);

title(['PLL DSB IPN(Band = L7)']);hold on;
semilogx(f,So_ref_dB,'k-.','LineWidth',2);
semilogx(f,So_ref_dB+So_refgen_dB,'k-.','LineWidth',2);
semilogx(f,So_vco_dB,'g--','LineWidth',2);
semilogx(f,So_dsm_dB,'m--','LineWidth',2);
semilogx(f,So_cp_dB,'b-.','LineWidth',2);
semilogx(f,So_lpf_dB,'c-.','LineWidth',2);

legend(sprintf('tot (IPN = %0.3g dBc)',10*log10(IPN_tot)),
...
sprintf('REF (IPN = %0.3g dBc)',10*log10(IPN_ref+IPN_refgen)), ...
sprintf('VCO (IPN = %0.3g dBc)',10*log10(IPN_vco)), ...
sprintf('DSM (IPN = %0.3g dBc)',10*log10(IPN_dsm)), ...
sprintf('CP (IPN = %0.3g dBc)',10*log10(IPN_cp)), ...
sprintf('LPF (IPN = %0.3g dBc)',10*log10(IPN_lpf)),
'Location','NorthEast');
```



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- [4] Woogeun Rhee, Phase-Locked Frequency Generation and Clocking, IET 2020.

